SupportPearlz — Technical Report (Working Draft)

AAI–407 / PRAC–03 | LangChain RAG Customer Support Agent

1. Problem and objective

Pearlz Home Systems receives recurring customer questions whose answers are distributed across product, warranty, returns, shipping, service and privacy documents. SupportPearlz is a retrieval-augmented support assistant designed to reduce document-search time while preferring an honest refusal over an unsupported answer. All company data and contact details in this project are fictional.

2. Architecture and design

The application separates offline ingestion from online querying. Ingestion dispatches files to loaders for Markdown, CSV, PDF, DOCX and TXT, attaches source metadata, removes duplicate content, applies recursive character splitting, embeds chunks using text-embedding-3-small and persists them in Chroma. The query path condenses follow-up questions, retrieves relevant chunks, applies a configurable relevance gate, and composes a LangChain prompt with gpt-4o-mini. Pydantic validates the structured answer and returned source labels are checked against labels supplied to the model.

3. Chunking experiment

The baseline is 800 characters with 120-character overlap. The assignment requires six measured configurations. The following table is intentionally unfilled until the student runs the experiment; no performance numbers have been invented.

Chunk size	Overlap	Chunk count	Mean / max length	Answer-containing top-4
300	0	To measure	To measure	To measure
300	60	To measure	To measure	To measure
800	0	To measure	To measure	To measure
800	120	To measure	To measure	To measure
1500	0	To measure	To measure	To measure
1500	225	To measure	To measure	To measure

Smaller chunks may improve precision but can remove qualifiers and context. Larger chunks preserve context but can dilute similarity ranking. The adopted configuration must be justified from measured top-4 results, not tutorial defaults.

4. Embeddings and vector store

The same embedding model is used for indexing and query-time embedding because vectors from different embedding models do not share a compatible semantic space. Chroma stores vectors and metadata locally. The index builder creates the index once; later runs load the persistent collection unless an explicit rebuild is requested. The current prototype supports full rebuild; a production incremental update should delete old chunks by source and upsert the revised document.

5. Retrieval comparison

The assignment asks for similarity search with k=3, similarity with k=8, and maximal marginal relevance (MMR), evaluated on the same labelled questions. The supplied evaluator currently measures expected-source retrieval hits. It does not yet implement the full MMR comparison table or answer-level judge metrics. These must be run and added before final submission.

6. Prompt design and safety

The final prompt specifies source-only grounding, refusal when context is absent, partial answers when only part is supported, source labels, concise customer language and resistance to instructions embedded in user text or retrieved documents. The assistant must not invent prices, warranty periods, delivery exceptions, compensation or actions. A weak-prompt versus final-prompt comparison still needs actual model outputs.

7. Evaluation and failure analysis

The project includes 20 evaluation cases across factual, multi-document, follow-up, unanswerable, ambiguous, adversarial and robustness categories. Run the index and evaluation harness, inspect each retrieved source and record pass/partial/fail verdicts. The included script reports retrieval-source hits only; it does not claim answer correctness, groundedness, refusal accuracy or citation precision. No success rate is claimed in this draft.

8. Security and deployment

Passwords are stored as salted PBKDF2 hashes. Registration creates standard users only; administrator provisioning is controlled through environment variables. The OpenAI key is validated at runtime and held in Streamlit session state rather than saved to source files or the user database. Docker volumes preserve the SQLite user database and Chroma index. The prototype still requires production security review, HTTPS, rate limiting, account recovery, session hardening and privacy review before use with real customers.

9. Limitations and remaining submission work

The generated knowledge base is a compact fictional dataset and is currently below the assignment target of approximately 9,000–12,000 words. The PDF product manual has 12 pages, and source formats include Markdown, CSV, DOCX and PDF. Before final submission, expand the knowledge corpus to the required word count, complete measured chunking and retrieval experiments, capture six genuine screenshots, run the live evaluation, attribute failures, and record a 5–8-minute demonstration. Do not submit placeholder values as measured evidence.

This PDF is a draft artifact for editing, not a claim that the remaining empirical work has been completed.